PORTFOLIO PROJECT · R · STATISTICS

What shapes a developer's pay?

Cleaning and testing a global developer survey in R: pay, experience, remote work and job satisfaction

Dhruv Khanna · MS Business Analytics, Quinnipiac University

65,437 responses · 48 variables

DATA PREPARATION

From raw survey to analysis-ready table

Filter

Dropped answers with zero or negative pay or satisfaction.

Standardize

"Less than 1 year" became 0 and "More than 50 years" became 50, so experience is numeric.

Reshape

Split multi-select employment answers into yes/no columns. Typed categories as factors.

Missing data: 40 to 60% on the optional AI-opinion questions, 20 to 35% on pay and experience. Tests use about 15,900 complete responses.

THE TYPICAL RESPONDENT

An experienced, fairly satisfied developer

$66k

median total pay. The mean is $86k, pulled up by outliers.

12

median years of coding

8

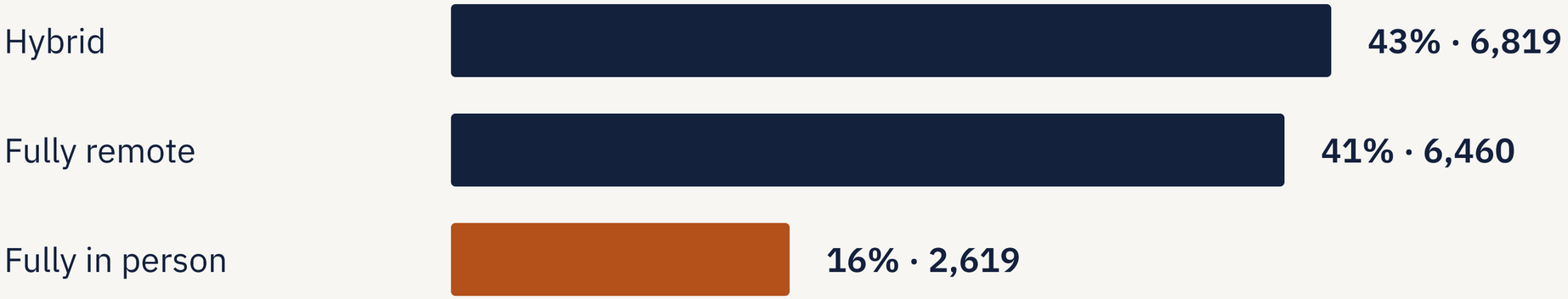
median years coding professionally

7.0

average job satisfaction out of 10

HOW DEVELOPERS WORK

Only one in six works fully in person



Remote setup varies strongly by country ($\chi^2 = 3{,}402.5$, $p < .001$), and pay differs across setups ($F = 35.3$, $p < .001$).

HYPOTHESIS TESTS

Seven tests, seven significant results

Test	Result	Reading
Chi-square: education × purchase influence	$\chi^2 = 114.0$, $p < .001$	Associated
Chi-square: country × remote setup	$\chi^2 = 3{,}402.5$, $p < .001$	Strongly associated
ANOVA: pay by remote setup	$F = 35.3$, $p < .001$	Means differ
ANOVA: satisfaction by education	$F = 2.35$, $p = .021$	Means differ slightly
Correlation: years coding × pay	$r = 0.15$	Weak positive
Correlation: pro years × satisfaction	$r = 0.10$, $p < .001$	Weak positive
Correlation: years coding × satisfaction	$r = 0.095$, $p < .001$	Weak positive

THE REAL FINDING

With 15,900 responses, almost anything is significant. Size is what matters.

2% of the variation in pay is explained by years of coding. A correlation of 0.15 is real, but experience alone barely predicts what a developer earns.

NEXT STEPS

What I'd do with another week

Control for country

Remote work and pay both vary by country, so the pay gap by setup may be a country effect.

Model log pay

Pay is heavily skewed. A log scale stops a few huge salaries from steering the results.

One model, not seven tests

A multiple regression would show each factor's effect while holding the others fixed.